Can an LLM Estimate the Embodied Carbon of a Building Enclosure?

Error Analysis and Prompt Design for Early-Stage A1–A3 Estimates on 292 North American Buildings

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Abstract

We test whether a large language model (Claude Opus 5.5, run as a Claude Code subagent with medium reasoning effort) can estimate the cradle-to-gate (A1–A3) embodied carbon of the building enclosure from early-stage design features. The reference values come from the 292 buildings of the harmonized North American WBLCA dataset of Benke et al. [1], normalized per m^2 of floor area. With a plain prompt, only 19.5% of estimates fall within ±20% of the reference LCA. An error analysis shows that the model compresses all buildings toward a typical value (log–log slope 0.46), never considers HFC-blown XPS insulation (15.5% of all enclosure carbon, and about half of the systematic underestimate), underestimates envelope area when it is not given, and misjudges renovations. Its largest overestimates trace back to incomplete reference LCAs. A first revised prompt that added qualitative material knowledge and a step-by-step method made the model more consistent but *less* accurate: the model repeated the knowledge without applying it to the numbers. A second revision placed externally sourced quantitative anchors (EPD and CLF baseline intensities) at the step where each number is chosen. It raised coverage to 25.0%, lowered the bias from 0.75 to 0.86 (estimate/reference, geometric mean), and lowered the run-to-run CV from 19.0% to 12.5%. On a held-out test set of 88 buildings the gains point the same way but are not statistically significant. A feature study shows that ten early-design inputs are as informative as all 48; the binding constraint is missing assembly information, not prompt length or feature count. All data, prompts, code and model outputs are public.[1]

1 Introduction

Early design decisions fix most of a building's embodied carbon, and the exterior enclosure is one of the most consequential and variable of them. Early-stage LCA is hard because design information is sparse, and the usual hand-off from architect to LCA practitioner is slow. Recent work shows that LLM in-context learning can support early-stage whole-building LCA, with domain knowledge and ontologies improving on innate knowledge [2]. We narrow that question to the enclosure and to the product stage (A1–A3), where material choices dominate and where LEED v5 embodied-carbon credits are earned in practice. We ask:

Q1 How accurately and consistently does an LLM estimate enclosure A1–A3 embodied carbon from early-stage features, and why does it fail?

[1] https://github.com/HeshanChen/CEE226LCA

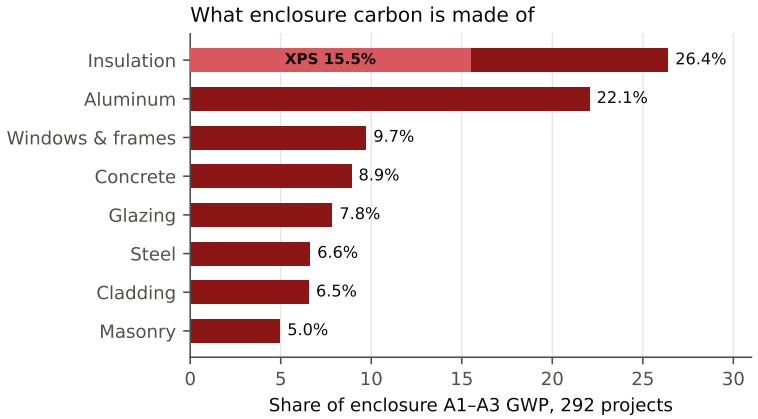


Figure 1: Share of enclosure A1–A3 GWP by material group across the 292 projects. XPS alone is 15.5%, at 43.6 $kgCO_2e/kg$, against 1.2 for mineral wool and 3.1 for polyiso. This is consistent with the HFC-blown XPS modelled by Tally and One Click LCA in the assessment years. 149 projects contain XPS.

Q2 Which prompt changes improve the estimates, and which do not?

Q3 Which input features does the model actually need?

2 Data and Ground Truth

Source. The harmonized dataset of Benke et al. [1] (CC BY 4.0) contains 927,950 material-level LCA results for 292 projects in the US and Canada, with 48 early-stage features per project (49 columns including the project index).

Scope and unit. We keep OmniClass element `Shell - Enclosure` and stage `A1-A3` (55,807 rows) and divide the total by floor area. We use `bldg_cfa` for new construction and `bldg_renovated_gfa + bldg_added_gfa` for the 49 renovations, which reproduces the dataset's own `eci_a_to_c_cfa` normalization for all 292 projects. The enclosure includes windows, doors and roofs, and the unit is per m^2 of *floor* area. This differs from the proposal's functional unit (1 m^2 of wall), because wall area is missing for most projects.

Distribution. The reference values have a median of 78.0 $kgCO_2e/m^2$, a mean of 95.6 and a range of 0.49–513.9 (5th–95th percentile 10.5–255.7). The enclosure is a median 19% of whole-building A–C carbon.

Composition. Insulation (26.4%) and aluminum (22.0%) dominate the enclosure total (Fig. 1).

Data quality. Ten new-construction projects have no window, glazing, curtain-wall or storefront GWP at all, although several report a window-to-wall ratio of 0.5–0.7. Project 74, for example, has WWR 0.65 and a reference value of 0.64 $kgCO_2e/m^2$. We keep these projects in every headline metric and also report results without them ("excl. incomplete").

3 Methods

Model and isolation. Each estimate is produced by a Claude Code subagent (Claude Opus 5.5, reasoning effort medium) that may read only its task file and write only its answer files. It has no web access, no code execution and no access to the ground truth. A task file contains the system prompt, a shared context (the feature glossary) and 48–49 building blocks, each with the building's non-empty features followed by

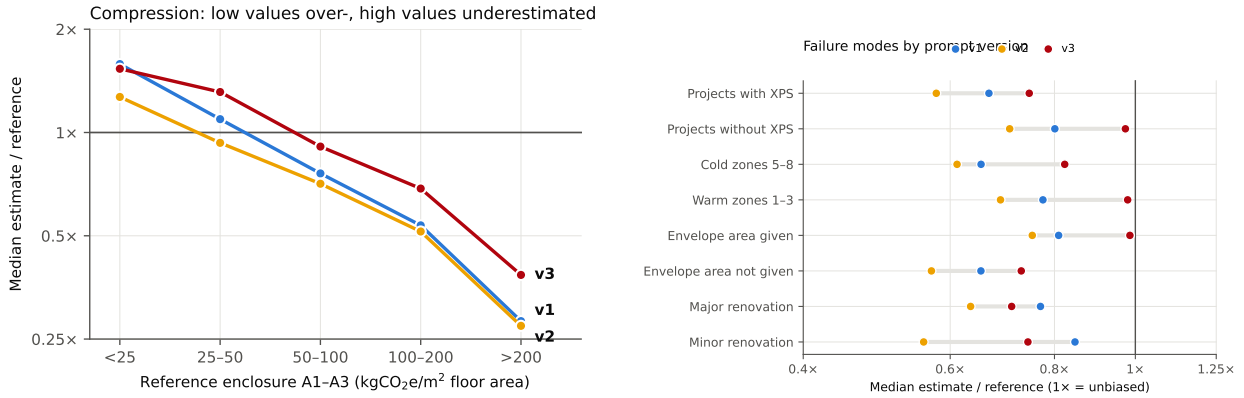


Figure 2: Left: median estimate/reference by reference-value bin. All prompts compress toward a typical value; v3 shifts the curve up. Right: median estimate/reference for the subgroups behind each failure mode.

the task text. Appendix A gives the exact text. The model is told to treat every building as an independent request.

Protocol. Each prompt version is run 3 times on all 292 buildings (876 estimates). For every run, the buildings are shuffled with a fixed seed (1000+ run index) and split into six chunks. The chunks are identical across prompt versions, so results can be paired building by building. The estimate for a building is the mean of the midpoints of its returned ranges. In one v3 run a container restart interrupted the subagents after 130 answers; the remaining 162 buildings were answered by continuation tasks with the same prompt.

Parsing. We use the course parser (the last line `RANGE_kgCO2e_per_m2: low - high`) with one change: a lower bound of 0 is accepted. Renovations that leave the envelope untouched have reference values near zero, and the original parser would have discarded exactly those answers.

Metrics.

- Accuracy: the share of buildings whose reference value lies within the estimate ±20% (coverage, with Wilson 95% CI), RMSE, median absolute percentage error (APE), and the median $|\log(\text{estimate}/\text{reference})|$.
- Bias: the geometric mean of estimate/reference.
- Consistency: the mean coefficient of variation (CV) across runs.
- Paired comparisons between prompt versions: McNemar's test on coverage and the Wilcoxon signed-rank test on $|\log(\text{estimate}/\text{reference})|$.

Held-out split. No independent dataset with the same structure exists (Section 6). We therefore split the 292 projects into a dev set of 204 and a test set of 88 (seed 226), stratified on input features only: new construction vs. renovation × climate group.

4 Baseline and Error Analysis

With prompt v1 (Appendix A), all 876 answers parsed.

- Coverage is 19.5% (CI 15.4–24.5%), RMSE 79.7 $kgCO_2e/m^2$, median APE 37.9%, bias 0.75 and CV 19.0% (Table 1).
- 63% of buildings are underestimated beyond the ±20% band, and 17.5% overestimated.

Compression, not uniform underestimation.

- The log–log slope of estimate on reference is 0.46: buildings below 25 $kgCO_2e/m^2$ are overestimated 1.58×, and buildings above 200 are estimated at 0.28× (Fig. 2).
- The estimates span 1.7–168 $kgCO_2e/m^2$, against 0.5–514 for the reference.
- The 20 largest errors, all underestimates, contribute 68% of the squared error.
- The model leans on geometry. The envelope-to-floor ratio correlates 0.62 with the log estimate but only 0.39 with the log reference. What it cannot predict is the carbon intensity per m^2 of envelope, which correlates -0.79 with the log error. That intensity is set by assembly and material choices, and *none of the 48 features describes the wall assembly*.

Why the model underestimates.

1. *HFC-blown XPS is never considered.* XPS appears in 149 projects, yet only 0.6% of answers mention it and none mention blowing agents. Median estimate/reference is 0.67 with XPS and 0.80 without. Removing XPS from the reference raises the median ratio from 0.72 to 0.84, so XPS explains about half of the bias.
2. *Climate is an insulation effect.* The ratio is 0.65 in cold zones (5–8) and 0.77 in warm zones (1–3). XPS is present in 65–68% of zone 5A/5B projects but only 26–33% of zone 3B/3C projects.
3. *Missing envelope area.* The ratio is 0.81 when `bldg_therm_env_area` is given and 0.65 when it is not; the model's own take-off is too small.
4. *Curtain-wall tail.* Single projects reach 300 $kgCO_2e/m^2$ from aluminum curtain wall alone.
5. *Renovations scoped too narrowly.* 60% of major renovations are underestimated, and tenant improvements reach 0.24.
6. *Scope wording.* The v1 scope text was narrower than the dataset's enclosure. For example, 11% of enclosure GWP was tagged "Superstructure" in the source tool, mostly roof insulation and membranes.

Why the model overestimates. The ten flagged incomplete references have a median ratio of 3.09; here the model's estimate is a plausible building and the reference is not. The remaining overestimates are mostly tenant improvements and minor renovations, where the features do not reveal how much envelope was replaced.

Ruled out. R-values, WWR, size, completion year and the embodied-carbon-reduction flag each correlate weakly with the error ($|r| \leq 0.26$).

5 Prompt Iterations

5.1 Prompt v2: qualitative knowledge and a step-by-step method

Prompt v2 addressed each finding:

- a scope written to match the dataset's enclosure;
- a required component take-off (areas → assemblies → intensities → total);
- named high-impact materials: HFC-blown XPS at "roughly ten times or more" the GWP of other insulation per unit of thermal resistance [4, 5], and aluminum curtain wall at 100–200 $kgCO_2e/m^2$ of facade [6, 7];
- an instruction not to anchor on a typical building;
- guidance for renovations;

Table 1: Accuracy and consistency by prompt version (3 runs each). Coverage: reference within estimate ±20%. Bias: geometric mean of estimate/reference.

Buildings	Prompt	Coverage (95% CI)	RMSE	Median APE	Med. $\lvert\log\rvert$	Bias	CV
All (292)	v1	19.5% (15.4–24.5)	79.7	**37.9%**	0.456	0.75	19.0%
	v2	15.1% (11.4–19.6)	81.0	44.3%	0.563	0.67	13.8%
	v3	**25.0%** (20.4–30.3)	**76.0**	40.0%	**0.426**	**0.86**	**12.5%**
New constr. (243)	v1	18.9%	77.4	37.5%	–	0.76	17.3%
	v3	**26.3%**	**74.1**	38.0%	–	**0.91**	**10.9%**
Test (88)	v1	20.5% (13.3–30.0)	74.1	**36.4%**	**0.393**	0.75	19.2%
	v2	18.2% (11.5–27.5)	75.7	43.4%	0.511	0.66	15.2%
	v3	**26.1%** (18.1–36.2)	**69.2**	37.0%	0.400	**0.86**	**11.2%**

- a `BEST` line in addition to the range.

Result: more consistent, less accurate.

- v2 lowered the CV from 19.0% to 13.8% and raised the log–log slope to 0.55.
- But coverage fell to 15.1% and the bias to 0.67 (Table 1). The errors were larger (Wilcoxon $p < 0.0001$), and v2 was closer to the reference for only 34% of buildings.

Three mechanisms explain this:

1. *Knowing is not applying.* v2 answers mention XPS 50 times more often, but 89% still assume polyiso roof insulation and place XPS below grade. The median roof intensity stays at 35 $kgCO_2e/m^2$, and projects with XPS fall from 0.67 to 0.58.
2. *Low bottom-up intensities.* Opaque walls were taken at 80, roofs at 35 and curtain wall at 145 $kgCO_2e/m^2$, giving 66.8 per m^2 of envelope against 86.7 in the reference. The prompt's curtain-wall range became the default value.
3. *A rule that subtracts.* "Count only new materials" pushed minor renovations from 0.85 to 0.56.

The self-reported `BEST` value sat below the range midpoint and was less accurate.

5.2 Prompt v3: quantitative anchors at the decision step

v3 keeps the v2 scope, method and anti-anchoring rule and changes four things (Appendix A.6):

1. A reference table of A1–A3 intensities per m^2 of component, compiled by a separate research sub-agent from external sources, explicitly excluding the ground-truth dataset and the CLF WBLCA benchmark. The sources are CLF North American Material Baselines 2023/2025, PCI precast EPDs, Kawneer fenestration EPDs, RDH wall studies and others [3, 9, 8, 10]. The table covers wall types, fenestration, roof assemblies by insulation type (HFC XPS: 85–110 counting the product stage only, up to 335–410 when blowing-agent emissions are booked with the product), insulation per $m^2 \cdot$RSI and aluminum per kg.
2. The tool modelling convention: LCA models of this period often model roof and below-grade insulation as HFC XPS, so the range must include that case when the insulation type is unknown.
3. A per-envelope sanity check, with a threshold of about 50 $kgCO_2e/m^2$ of envelope derived from the component ranges.
4. A renovation rule that no longer subtracts, and a single `RANGE` output.

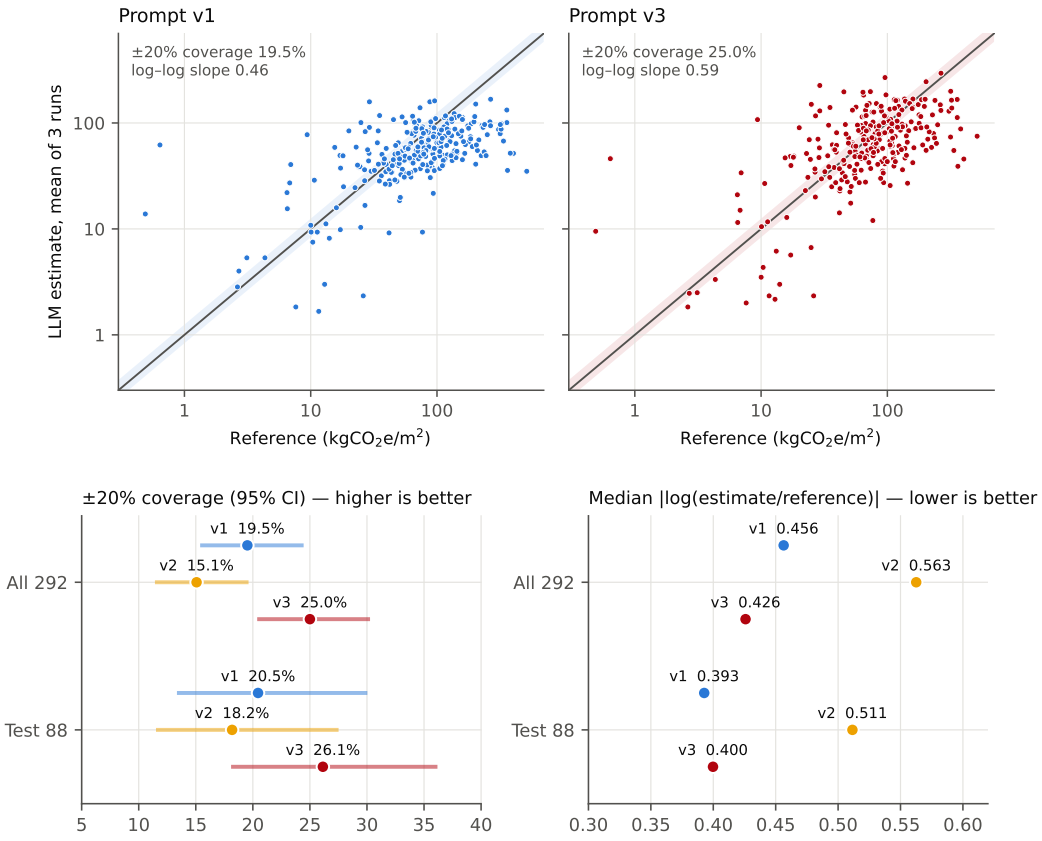


Figure 3: Top: estimate vs. reference for v1 and v3 (line 1:1, band ±20%). Bottom: coverage with 95% CI and median $|\log(\text{estimate}/\text{reference})|$ for all 292 buildings and the held-out 88.

Result.

- v3 is the best prompt on coverage (25.0%), RMSE (76.0), log error (0.426), bias (0.86) and consistency (CV 12.5%) (Table 1, Fig. 3).
- Against v2 the gain is significant (McNemar $p = 0.002$; Wilcoxon $p < 0.0001$). Against v1 it is consistent but borderline (McNemar $p = 0.085$; Wilcoxon $p = 0.053$; v3 closer for 57% of buildings).
- 96% of answers for XPS projects now consider HFC XPS.
- The non-XPS part of the enclosure is unbiased: estimate/(reference − XPS) went from 0.84 to 0.99. Buildings with a given envelope area are also unbiased (0.99), and warm zones reach 0.98.
- What remains: compression at the top (0.38 above 200 $\text{kgCO}_2\text{e/m}^2$); more overestimation of light enclosures (1.31 in the 25–50 bin); 0.73 when the envelope area is missing; and renovations (0.71–0.74).
- The median APE is slightly above v1 (40.0% vs. 37.9%), because v3 shifts estimates up and APE penalizes overestimates more than underestimates.

Table 2: Feature study (prompt v3, run 0, 292 buildings, paired with all 48 features). Run-to-run spread of v3 itself: 2.0 points of coverage and 0.025 in median $|\log|$.

Condition	Features	Coverage	Med. $\lvert\log\rvert$	Δ Med. $\lvert\log\rvert$	Wilcoxon p	Bias
all	48	24.3%	0.461	–	–	0.86
early	39	21.6%	0.462	+0.001	0.13	0.88
env	15	19.9%	0.463	+0.001	0.33	0.93
min	10	25.3%	0.476	+0.015	0.81	0.81
no-lca	40	25.3%	0.441	−0.020	0.57	0.90
no-mass	47	22.6%	0.452	−0.009	0.92	0.85

6 Held-out Evaluation

No external dataset. The CLF WBLCA Benchmark v2 probably shares source models with the ground truth. The open CarbEnMats database [11] reports envelope A1–A3 for only 172 European buildings, uses a different element definition, and has no geometry fields for them. We therefore hold out 88 test buildings.

Post hoc split. The split was assigned after the runs, and the error analysis had seen aggregate patterns across all 292 buildings. To check that this did not drive the prompt design, we re-ran every diagnostic on the 204 dev buildings alone. Every finding replicates, including:

- the log–log slope (0.44 vs. 0.46);
- the XPS share (14.7% vs. 15.5%) and the XPS effect (0.66 vs. 0.77 with and without XPS);
- the envelope-area effect (0.79 vs. 0.65 with and without it);
- v2's polyiso assumption (89%) and its renovation drop ($0.79 \rightarrow 0.40$).

The same prompt revisions would have followed from the dev set alone.

Test results.

- On the 88 test buildings, v3 improves on v1 in coverage (26.1% vs. 20.5%), RMSE (69.2 vs. 74.1), bias (0.86 vs. 0.75) and CV (11.2% vs. 19.2%), but its median log error is unchanged (0.400 vs. 0.393).
- None of the v1–v3 differences is significant at this size (McNemar $p = 0.41$; Wilcoxon $p = 0.38$).
- v3 is significantly better than v2 (Wilcoxon $p = 0.002$), and v2's regression against v1 replicates ($p = 0.0002$).

7 Feature Study

With v3 fixed, we varied the input features and ran each condition once (run 0), pairing it with run 0 of v3 on all 48 features. The conditions were:

- *early*: 39 features; no LCA-derived fields and no `mass_total`, which is the reference LCA's own inventory mass and therefore unknown at design time;
- *env*: 15 envelope-relevant features;
- *min*: 10 features (climate zone, use, project type, floor area, storeys, WWR, R-values);
- *no-lca*: all features except the 8 LCA-derived fields;
- *no-mass*: all features except `mass_total`.

Results (Table 2).

- No feature set differs significantly from the full set, and every difference lies within the run-to-run spread of v3 itself.
- Ten early-design inputs are as good as 48.
- The model does not depend on `mass_total` or the other LCA-derived fields.

Prompt content moved the median $|\log|$ error by 0.137 from v2 to v3; no feature set moved it by more than 0.020. The missing information is the assembly type, which none of the features provides.

8 Discussion

Prompt design. Three lessons transfer beyond this dataset.

1. A structured, step-by-step method improves consistency, but it also passes the model's component-level priors straight into the total.
2. Qualitative knowledge in a prompt is repeated in the reasoning but does not move the numbers. Quantitative, sourced anchors placed at the step where a number is chosen, with the condition under which they apply, do move them.
3. Rules that subtract (exclusions, "only new materials") amplify an existing low bias unless they are paired with what to include.

Early design and LEED v5. An enclosure estimate within $\pm 20\%$ of a full LCA is reached for about a quarter of buildings, with a bias of -14%. That is useful for comparing assemblies and spotting hotspots (curtain wall, HFC XPS, insulated metal panels) at schematic design. It is not reliable enough to predict a credit outcome for an individual building. The largest gains in accuracy would come from adding the assembly type to the inputs, which the proposal anticipated, rather than from further prompt engineering.

9 Limitations

- Answers within a task file share a context, so they are less independent than separate API calls.
- The held-out split was assigned after the runs; the dev-only replication mitigates but does not replace an external test.
- With 3 runs, the v3–v1 gain is borderline significant on all 292 buildings and not significant on the test set.
- The feature study used one run per condition, and 7 of 9 planned leave-one-group-out ablations were stopped to limit cost, so it is a null result, not proof of irrelevance.
- The unit is per m^2 of floor area, and the enclosure includes windows, doors and roofs.
- Reference quality is uneven: there are incomplete inventories, and the tools' data vintages differ (HFC XPS).
- Results are specific to Claude Opus 5.5 at medium effort.

10 Conclusion

A frontier LLM can produce enclosure A1–A3 estimates from early-stage features. Its main error is compression toward a typical value, together with blind spots for specific high-carbon materials. Error analysis on the model's own reasoning texts located those blind spots. Placing sourced quantitative anchors at the decision step reduced the bias from 0.75 to 0.86 and the run-to-run variation by a third, and raised coverage from 19.5% to 25.0%. Further progress depends on giving the model the assembly information that early-stage feature sets currently lack.

References

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A Prompts

The prompts below are exported verbatim from `experiment_config.py` by `report/latex/make_prompts.py`. Six non-ASCII characters in the glossary are shown as ASCII (e.g. CO2 for CO_2). Each request consists of the system prompt, the shared context, and a building block (the building's features followed by the task text).

A.1 System prompt

```
You are a building LCA expert.
```

A.2 Shared context (identical in v1, v2 and v3)

```
Forget the information from previous sessions and calculate everything from the beginning.

data_glossary_LLM contains feature names and descriptions for the data:

- project_index: Unique project index number
- site_country: Country where the project is located
- site_region: Region where the project is located. Based on the 2019 National Ready Mix Concrete
    Association regions.
- site_state_province: State or province where the project is located.
- site_clim_zone: Climate Zone where the project is located per respective energy code of country.
- bldg_proj_type: Type of construction which details whether the project is new or a type of modification
    to an existing building.
- bldg_compl_year: Year of project's actual construction completion, certificate of occupancy, or
    anticipated project completion year.
- bldg_code_year: Corresponding year of the international building code (IBC) or other equivalent the
    project complied with.
- bldg_energy_code_year: Corresponding year of the international energy conservation code (IECC) or other
    equivalent the project complied with.
- bldg_ibc_type: Type of construction per IBC.
- bldg_park_type: Type of attached/integrated parking structure or "parkade".
- bldg_cfa [m2]: Total constructed floor area (CFA) which includes the gross internal floor area (GFA) of
    the building and the gross floor area of any attached/integrated parking components.
- bldg_gfa [m2]: Total gross internal floor area (GFA) of the building.
- bldg_parking_gfa [m2]: The total horizontal area occupied by any attached/integrated parking components.
- bldg_added_gfa [m2]: The total horizontal area where new floor area was added to an existing building
    (i.e., the floor area of an addition).
- bldg_renovated_gfa [m2]: The total horizontal area where the existing building was renovated.
- bldg_prim_use: Primary classification of the building by principal activity
- bldg_sec_use: Secondary classification of building by principal activity.
- bldg_occupants: Total building occupancy as derived from building code
- bldg_res_units: Total number of residential units in the building
- bldg_stories_above: Binned categorization of the number of stories above finished ground level.
- bldg_stories_below: Binned categorization of the number of stories below finished ground level.
- bldg_height [m]: Binned categorization of the total height of the building above finished ground level.
- bldg_therm_env_area [m2]: Total area of the building's thermal envelope.
- bldg_wwr: The ratio of total vertical fenestration area to total above-grade wall area
- bldg_rval_walls: Average nominal R-value of building's above-grade walls.
- bldg_rval_roofs: Average nominal R-value of building's roofs.
- str_seis_site_cls: Classification of the site for seismic design based on its soil and engineering
    properties.
- str_sdc: Categorization of required seismic design based on risk category and site location.
- str_wind_speed [m/s]: Ultimate wind speed used for ultimate wind design based on location.
- str_prim_horiz_sys: Type of primary horizontal gravity system. Structural floor construction system that
    supports at least 2/3 of the combined floor and roof area of the superstructure.
- str_prim_vert_sys: Type of primary vertical gravity system. System that transfers gravity loads to the
    foundation of the building.
- str_lat_sys: Type of lateral system that resists lateral loads of the building.
- str_podium: Designation for whether the project is a podium building - a building with two distinct zones
    with two different structural materials and systems
- str_sec_horiz_sys: The horizontal gravity system (as defined in Primary Horizontal Gravity System) for
    the podium portion of the building.
- str_sec_vert_sys: The vertical gravity system (as defined in Primary Vertical Gravity System) for the
    podium portion of the building
- str_grid_long [m]: Average center to center spacing of longer typical project column grid.
- str_grid_short [m]: Average center to center spacing of shorter typical project column grid.
- str_fdn_type: Type of foundation used to support gravity and lateral loads of project.
- str_sys_summary: Simplified summary of the buildings primary horizontal, vertical, and lateral system.
- lca_assessment_year: Year the LCA was performed
- lca_design_phase: The design phase of the project when the LCA was performed
- lca_RSP [years]: Reference study period for the LCA
- lca_software: Software tool used for the LCA
- lca_purp_of_assessment: Primary reason the LCA was undertaken.
- lca_phys_scope: The physical scope (building elements) included in the assessment.
- lca_ec_reductions: Yes/No designation to indicate if the WBLCA results reflect efforts by the design or
    construction team to reduce the embodied carbon of the project during design, procurement, and/or
```

```
    construction of the building.
- lca_ec_reduction_percent: Approximate percent reduction in embodied carbon of the building compared to a
    baseline model, if applicable.
- mass_total [kg]: Total mass of project. Sum of the project's total mass (inv_mass) for life cycle stages
    A-C
- stored_carbon_total [kg CO2e]: Total stored carbon of project. Sum of the project's total stored carbon
    (inv_stored_carbon) for life cycle stages A1-A3
- gwp_a_to_c [kg CO2e]: Sum of the project's total global warming potential (gwp) for life cycle stages A-C
- ap_a_to_c [kg SO2e]: Sum of the project's total acidification potential (ap) for life cycle stages A-C
- ep_a_to_c [kg Ne]: Sum of the project's total eutrophication potential (ep) for life cycle stages A-C
- sfp_a_to_c [kg O3e]: Sum of the project's total smog formation potential (sfp) for life cycle stages A-C
- odp_a_to_c [kg CFC-11e]: Sum of the project's total ozone depletion potential (odp) for life cycle stages
    A-C
- nred_a_to_c [MJ]: Sum of the project's total non-renewable energy demand (nred) for life cycle stages A-C
- mui_total_GFA [kg/m3]: Total material use intensity. Sum of the project's total material use intensity
    (mui_gfa) over the full RSP normailized by GFA for new construction projects. Includes material
    replacements from module B.
- eci_a_to_c_GFA [kgCO2e/m2]: Embodied carbon intensity. Sum of the project's total global warming
    potential (gwp) for life cycle stages A-C normalized by GFA for new construction projects.
- epi_a_to_c_GFA [kgNe/m2]: Eutrophication potential intensity. Sum of the project's total eutrophication
    potential (ep) for life cycle stages A-C normalized by GFA for new construction projects.
- api_a_to_c_GFA [kgSO2e/m2]: Acidification potential intensity. Sum of the project's total acidification
    potential (ap) for life cycle stages A-C normalized by GFA for new construction projects.
- sfpi_a_to_c_GFA [kgO3e/m2]: Smog formation potential intensity. Sum of the project's total smog formation
    potential (sfp) for life cycle stages A-C normalized by GFA for new construction projects.
- odpi_a_to_c_GFA [kgCFC11e/m2]: Ozone depletion potential intensity. Sum of the project's total ozone
    depletion potential (odp) for life cycle stages A-C normalized by GFA for new construction projects.
- nredi_a_to_c_GFA [MJ/m2]: Non-renewable energy demand intensity. Sum of the project's total non-renewable
    energy depletion intensity (nredi) for life cycle stages A-C normalized by GFA for new construction
    projects.
- mui_total_CFA [kg/m2]: Total material use intensity. Sum of the project's total material use intensity
    (mui_cfa) over the full RSP normailized by CFA for new construction projects. Includes material
    replacements from module B.
- eci_a_to_c_CFA [kgCO2e/m2]: Embodied carbon intensity. Sum of the project's total global warming
    potential (gwp) for life cycle stages A-C normalized by CFA for new construction projects.
- epi_a_to_c_CFA [kgNe/m2]: Eutrophication potential intensity. Sum of the project's total eutrophication
    potential (ep) for life cycle stages A-C normalized by CFA for new construction projects.
- api_a_to_c_CFA [kgSO2e/m2]: Acidification potential intensity. Sum of the project's total acidification
    potential (ap) for life cycle stages A-C normalized by CFA for new construction projects.
- sfpi_a_to_c_CFA [kgO3e/m2]: Smog formation potential intensity. Sum of the project's total smog formation
    potential (sfp) for life cycle stages A-C normalized by CFA for new construction projects.
- odpi_a_to_c_CFA [kgCFC11e/m2]: Ozone depletion potential intensity. Sum of the project's total ozone
    depletion potential (odp) for life cycle stages A-C normalized by CFA for new construction projects.
- nredi_a_to_c_CFA [MJ/m2]: Non-renewable energy demand intensity. Sum of the project's total non-renewable
    energy depletion intensity (nredi) for life cycle stages A-C normalized by CFA for new construction
    projects.
- ec_per_occupant_a_to_c [
kgCO2e/occupant

]: Embodied carbon per occupant. Sum of the project's total global warming potential (gwp) for life cycle
    stages A-C normalized by the number of buildings occupants (bldg_occupants).
- ec_per_res_unit_a_to_c [kgCO2e/ unit]: Embodied carbon per residential unit. Summary of the project's
    total GWP (gwp) for life cycle stages A-C normalized by number of residential units (bldg_res_units).
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan [kg]: nan
- nan [kg CO2e]: nan
- nan [kg CO2e]: nan
- nan [kg SO2e]: nan
- nan [kg Ne]: nan
```

```
- nan [kg O3e]: nan
- nan [kg CFC-11e]: nan
- nan [MJ]: nan
- nan [kg/m2]: nan
- nan [kg/m2]: nan

Reference the given feature information above. Estimate the embodied carbon of the building enclosure of
     the building below:
```

A.3 Example building features (project 225, all 48 features)

```
site_country: United States
site_region: Eastern
site_clim_zone: Zone 5A (Cool Humid)
bldg_code_year: 2015
bldg_energy_code_year: 2009
bldg_proj_type: New Construction
bldg_ibc_type: Type I-A
bldg_park_type: No Parking
bldg_cfa: 25000
bldg_gfa: 25000
bldg_prim_use: Healthcare
bldg_occupants: 2062
bldg_stories_above: 6 to 10
bldg_stories_below: 2
bldg_height: 22.6-30 m
bldg_therm_env_area: 12133.1318
bldg_wwr: 0.4
str_seis_site_cls: D
str_sdc: C
str_wind_speed: 56.77408
str_prim_horiz_sys: Steel: Frame + concrete on metal deck
str_prim_vert_sys: Steel: Columns
str_lat_sys: Steel: Moment frames
str_podium: Primary system defined above is on a podium
str_sec_horiz_sys: Concrete: Non-PT framing
str_sec_vert_sys: Concrete: CIP
str_grid_long: 10.668
str_grid_short: 10.668
str_fdn_type: Deep foundation > 50' (15m)
str_sys_summary: Steel
lca_assessment_year: 2023
lca_design_phase: Construction
lca_rsp: 60
lca_software: Tally LCA
lca_purp_of_assessment: Voluntary: Internal Benchmarking
lca_phys_scope: BSECF
lca_ec_reductions: Yes
lca_ec_reduction_percent: 0.106
mass_total: 39350375.72180476
```

A.4 Task text v1 (baseline)

```
Scope: the building enclosure only, i.e. OmniClass "Shell - Enclosure" (exterior walls and cladding,
     exterior windows, curtain walls and storefronts, exterior doors, roofing, and their insulation).
     Exclude substructure, superstructure, interior construction and interior finishes. Life cycle stages:
     A1-A3 only (product stage: raw material supply, transport to the factory, manufacturing). Exclude A4,
     A5, B, C and D.

Provide a plausible range of the enclosure's A1-A3 embodied carbon in units of kgCO2e per m2 of building
     floor area (a range, not a single value). For floor area, use bldg_cfa. If bldg_cfa is not provided,
     use bldg_renovated_gfa + bldg_added_gfa. Briefly state your method and key assumptions.
```

```
End your answer with exactly one final line in this format and nothing after it:
RANGE_kgCO2e_per_m2: <low> - <high>
(plain numbers only, no units or other text on that line)
```

A.5 Task text v2

```
Scope. Estimate the embodied carbon of the building enclosure as classified in this dataset (OmniClass
     "Shell - Enclosure"), life cycle stages A1-A3 only (raw material supply, transport to the factory,
     manufacturing). Exclude A4, A5, B, C and D. The enclosure includes:
- exterior walls as complete assemblies: cladding, backup framing or studs, sheathing, cavity and
     continuous insulation, air and vapor barriers, and the interior gypsum layer of exterior walls;
     structural exterior walls (concrete, CMU, tilt-up) when they form the wall;
- fenestration: windows, curtain wall, storefront and skylights (frames and glass), and exterior doors;
- the roof assembly above the structural deck: insulation, cover board, membrane or roofing;
- insulation at slab edges, below grade and under slabs that belongs to the thermal envelope.
It excludes foundations, the structural frame, floor and roof decks or slabs, interior partitions and
     interior finishes.

Method. Work through these steps and show the key numbers:
1. Areas. If bldg_therm_env_area is given, use it as the total envelope area. Otherwise estimate the roof
     area (footprint = floor area / stories above grade), the above-grade wall area (perimeter x height)
     and the glazed area (bldg_wwr x wall area), plus any insulated ground-floor or below-grade area. State
     the resulting envelope-to-floor-area ratio; single-storey and low-rise buildings often exceed 1.
2. Assemblies. From the climate zone, R-values, building use, structure, region and construction type,
     choose the most likely wall, glazing and roof assemblies. Name the insulation type and estimate its
     thickness from the R-value.
3. Intensities. Give each component an A1-A3 intensity in kgCO2e per m2 of that component, using values
     typical of the North American whole-building LCA tools of the assessment year (Tally, One Click LCA).
     Account for these high-impact materials:
   - XPS insulation made with HFC blowing agents is common in these tools and carries roughly ten times or
     more the GWP of mineral wool, fiberglass or polyiso per unit of thermal resistance, so a thick XPS
     layer (for example roof or below-grade insulation in cold climates) can dominate the enclosure total.
   - Aluminum is carbon-intensive per kg (curtain wall and storefront framing, window frames, metal panels,
     ACM). Aluminum-framed curtain wall is about 100-200 kgCO2e per m2 of facade, and fully glazed curtain
     wall facades are among the highest-carbon enclosures.
   - Insulated glass units, precast concrete panels and masonry are also significant.
4. Total. Sum the components, divide by the floor area, and give a range that reflects your uncertainty
     about the assemblies. Do not pull the estimate toward a typical building: metal-clad, curtain-wall or
     thickly insulated enclosures can exceed 200 kgCO2e per m2 of floor area, and projects with little new
     enclosure work can fall below 10.
5. Renovations. For renovations and tenant improvements, count only new enclosure materials. Use
     bldg_proj_type, lca_phys_scope and mass_total per floor area to judge how much new envelope work was
     done. Major renovations often include new windows, cladding or roofing, and a high mass_total per
     floor area indicates substantial new material.

Floor area: use bldg_cfa. If bldg_cfa is not provided, use bldg_renovated_gfa + bldg_added_gfa.

End your answer with exactly two final lines in this format and nothing after them:
BEST_kgCO2e_per_m2: <value>
RANGE_kgCO2e_per_m2: <low> - <high>
(plain numbers only, no units or other text on these lines)
```

A.6 Task text v3 (final prompt)

The reference values are documented with sources, system boundaries and unit conversions in `references/envelope_intensities_sources.md`.

```
Scope. Estimate the embodied carbon of the building enclosure as classified in this dataset (OmniClass
     "Shell - Enclosure"), life cycle stages A1-A3 only (raw material supply, transport to the factory,
     manufacturing). Exclude A4, A5, B, C and D. The enclosure includes:
- exterior walls as complete assemblies: cladding, backup framing or studs, sheathing, cavity and
     continuous insulation, air and vapor barriers, and the interior gypsum layer of exterior walls;
     structural exterior walls (concrete, CMU, tilt-up) when they form the wall;
```

```
- fenestration: windows, curtain wall, storefront and skylights (frames and glass), and exterior doors;
- the roof assembly above the structural deck: insulation, cover board, membrane or roofing;
- insulation at slab edges, below grade and under slabs that belongs to the thermal envelope.
It excludes foundations, the structural frame, floor and roof decks or slabs, interior partitions and
     interior finishes.

Method. Work through these steps and show the key numbers:
1. Areas. If bldg_therm_env_area is given, use it as the total envelope area. Otherwise estimate the roof
     area (footprint = floor area / stories above grade), the above-grade wall area (perimeter x height)
     and the glazed area (bldg_wwr x wall area), plus any insulated ground-floor or below-grade area. State
     the resulting envelope-to-floor-area ratio; single-storey and low-rise buildings often exceed 1.
2. Assemblies. From the climate zone, R-values, building use, structure, region and construction type,
     choose the most likely wall, glazing and roof assemblies, and name the insulation type and thickness.
     Whole-building LCA models from this period (Tally, One Click LCA) often model roof and below-grade
     insulation as XPS made with HFC blowing agents. When the insulation type is not known, include that
     case in your range, especially in colder climates where the insulation is thick.
3. Intensities. Give each component an A1-A3 intensity in kgCO2e per m2 of that component. Use these
     reference values from EPDs and industry baselines (A1-A3, per m2 of component):
   Opaque walls (per m2 of wall, complete assembly):
   - steel stud + metal panel or ACM rainscreen: 70-110
   - steel stud + brick veneer: 90-120; brick veneer on CMU backup: 135-165
   - steel stud + EIFS: about 50; steel stud + fiber cement or stucco: 45-80
   - wood stud + vinyl or wood siding: 25-45; wood stud + brick or stone veneer: 60-100
   - insulated metal panel: newer products 27-72; legacy polyurethane (HFC) panels about 107 (2 in) to 274
     (6 in)
   - precast concrete sandwich panel: 90-165; architectural precast with backup wall: 80-130
   - tilt-up concrete with interior insulation and gypsum: 80-125
   - curtain-wall spandrel: 75-140
   Fenestration (per m2 of unit, frame and glass):
   - aluminum stick curtain wall: 120-150; aluminum unitized curtain wall: 150-210
   - aluminum storefront: 140-155; aluminum window wall: 170-180; aluminum windows: 95-140
   - vinyl windows: 40-90; fiberglass windows: 75-145
   Roof above the deck (per m2 of roof, about R-25 to R-30, with membrane and cover board):
   - polyiso: 20-28; mineral wool board: 35-55; low-GWP XPS: 28-40; metal standing seam over polyiso: 27-40
   - XPS with HFC blowing agent: 85-110 when only product-stage emissions are counted, and up to 335-410
     when the blowing-agent emissions are booked with the product, as some LCA tools do
   - insulated metal roof panel: 27-72 for newer products, about 190 for a legacy 4 in panel
   Insulation per m2 at RSI 1 (R-5.7): polyiso 2-4; EPS 2.5; mineral wool board 7-8; fiberglass batt about
     1; closed-cell spray foam 3-5 (HFO) or 20-36 (HFC); XPS 4-11 (low-GWP), about 18 (HFC, product stage
     only) or 58-99 (HFC, with blowing-agent emissions)
   Aluminum: about 6-13 kgCO2e per kg for North American extrusions; generic global aluminum data in LCA
     tools is about 16
Complete assemblies include framing, attachments, sheathing, barriers, sealants, flashing and trim, which
     early-stage take-offs tend to leave out; do not go below the low end of these ranges unless the
     assembly is clearly minimal.
4. Check. Convert your total to kgCO2e per m2 of envelope area and check that it is consistent with the
     reference values above. For new construction with conventional walls and 30-40% glazing, a blended
     value below about 50 kgCO2e per m2 of envelope usually means that a component, the aluminum and
     glazing content, or a high-carbon insulation option is missing; revisit those before you finalize.
5. Total. Divide the enclosure total by the floor area. Give a range that reflects your uncertainty about
     the assemblies, including the higher-carbon alternatives that are plausible for this building. Do not
     pull the estimate toward a typical building: metal-clad, curtain-wall or thickly insulated enclosures
     can exceed 200 kgCO2e per m2 of floor area, and projects with little new enclosure work can fall below
     10.
6. Renovations. For renovations and tenant improvements the LCA covers only the new work, but that work
     often includes substantial new envelope elements (re-cladding, new curtain wall or windows,
     re-roofing). Do not assume the existing envelope is kept unless the data indicates it. Use
     bldg_proj_type, lca_phys_scope and mass_total per floor area as evidence of how much new material was
     installed.

Floor area: use bldg_cfa. If bldg_cfa is not provided, use bldg_renovated_gfa + bldg_added_gfa.

End your answer with exactly one final line in this format and nothing after it:
RANGE_kgCO2e_per_m2: <low> - <high>
(plain numbers only, no units or other text on that line)
```

A.7 Subagent instructions (template)

Placeholders in braces are filled per task file by `subagent_runs.py prepare`.

```
You are the language model being tested in a building life-cycle-assessment (LCA) experiment for a
     university course project. Your job is to answer a set of estimation prompts using only your own
     knowledge.

Input file: {task_file}
It is about {n_lines:,} lines, so read all of it with the Read tool, using offset/limit to page past line
     2,000. It contains a System line, a shared context (a glossary of the feature names), and
     {n_buildings} building sections. Each section starts with a line like `===== KEY: {example_key}
     =====`, followed by the building's features and the question.

Rules. Breaking any of these invalidates the experiment:
- Read only that one task file. Do not open, list or search any other file or folder (especially nothing
     under results/, GWP-calculation/ or references/). Do not use web search or web fetch, and do not run
     code or shell commands. The only tools you may use are Read (on the task file) and Write (for your
     answer files).
- Treat each building as a separate, independent request, as if you were seeing it alone. Do not compare
     buildings, rank them, or adjust one estimate because of another.
- Follow each building's prompt exactly: {answer_length}, then end with {final_lines} containing plain
     numbers only.

Output: write your answers in parts of about 10 buildings each, using one new file per part and never
     overwriting an earlier part:
{answers_prefix}_part-1.md, then _part-2.md, and so on.
Inside each part file, every answer starts with its key line copied exactly from the task file, followed by
     the answer text, for example:

===== KEY: {example_key} =====
<answer>
{example_final}

Answer all {n_buildings} buildings, in the order they appear in the task file, with the key lines copied
     exactly.

When you are finished, reply with exactly one line: `done: <number of answers written> answers in <number
     of parts> parts`. Do not include the answers themselves in your reply.
```

B Reproduction

All steps are scripted in the repository:

```
python GWP-calculation/sum_gwp.py                  # ground truth (enclosure, A1-A3)
python GWP-calculation/enclosure_breakdown.py      # material breakdown + data-quality flags
python subagent_runs.py prepare --run-name 20261104_prompt-v3 --runs 3 --effort medium
#   one Claude Code subagent (Opus, effort medium) per results/<run>/subagent/tasks/*.instructions.txt
python subagent_runs.py resume  --run-name 20261104_prompt-v3   # only if interrupted
python subagent_runs.py collect --run-name 20261104_prompt-v3
python summarize_run.py 20261104_prompt-v3
python analysis/make_split.py
python analysis/compare_runs.py 20261104_baseline-innate 20261104_prompt-v2 20261104_prompt-v3
python analysis/error_analysis.py --split dev 20261104_baseline-innate 20261104_prompt-v2 20261104_prompt-v3
python analysis/feature_study.py
python report/make_figures.py
```